

Employee Performance Analysis

INX Future Inc | Multi-class Classification | Python, scikit-learn
Ishika • Data Science Portfolio Project

Problem Statement

INX Future Inc wanted to lift employee performance without hurting morale. The goal of this project is to find the factors that actually move performance and to build a model that classifies an employee's performance rating, so HR can spot risk early and act on evidence instead of intuition.

Dataset

- Employee-level records across departments: demographics, job role, satisfaction scores, experience and salary attributes.
- Target:** PerformanceRating — 2 (Low), 3 (Average), 4 (High).
- No missing values, no duplicates; the identifier column EmpNumber was dropped before modelling.
- Categorical fields label-encoded; class imbalance handled through macro-averaged metrics.

Exploratory Analysis

- Department:** average ratings vary clearly by department — role expectations and team environment matter.
- Satisfaction:** job satisfaction, environment satisfaction and work-life balance all track upward with performance.
- Correlation:** environment satisfaction and salary-hike variables show the strongest positive association with the target.

Top 3 Drivers of Performance

#	Factor	Why it matters
1	Last Salary Hike %	Strongest single predictor; pay recognition drives output.
2	Environment Satisfaction	Workplace environment shows a consistent, direct effect.
3	Years Since Last Promotion	Career stagnation is an early warning sign of decline.

Confirmed independently by EDA, correlation analysis and Random Forest feature importance.

Modelling

Framed as a multi-class classification task. Decision Tree and three Random Forest variants (baseline, improved, hyperparameter-tuned) were compared on accuracy and macro precision / recall / F1.

Model	Accuracy	Macro F1	Verdict
Decision Tree	~85%	~0.78	Overfits, unstable
Random Forest (baseline)	~91%	~0.84	Solid start
Random Forest (improved)	92%+	0.87	Final model
Random Forest (tuned)	~92%	~0.87	No real gain

Tuning added no meaningful lift, so the improved Random Forest was retained for simplicity and stability. The majority class is predicted with very high recall while minority classes stay usable — the model is positioned as a **decision-support tool, not an automated decision-maker**. It was serialised with `joblib`, reloaded and re-tested on sample employee records with identical output.

Business Recommendations

- Make salary hikes transparent and explicitly performance-linked.
- Invest in workplace environment and team culture — it is a measurable performance lever, not a soft one.
- Set structured promotion timelines to prevent stagnation.
- Run targeted training for employees flagged as low-performing.
- Use the model at hiring and review stages to assess performance risk early.

Conclusion

The project isolates the three factors that genuinely drive employee performance and delivers a Random Forest classifier that generalises well (92%+ accuracy, 0.87 macro F1). Together they give INX Future Inc a practical, data-driven route to improving performance while protecting morale.

Stack: Python · pandas · NumPy · scikit-learn · Matplotlib · Seaborn · joblib | **Author:** Ishika